Kavimalar D 241801118 Exp 07

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
```

```
df = pd.read_csv("/content/Mall_Customers.csv")

print("Dataset Loaded Successfully!")
```

```
Dataset Loaded Successfully!
```

```
print("First 5 Records:")
print(df.head())

print("\nLast 5 Records:")
print(df.tail())

print("\nShape of Dataset:")
print(df.shape)

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nNumber of Duplicate Records:")
print(df.duplicated().sum())

print("\nSummary Statistics:")
print(df.describe())
```

```
First 5 Records:
   CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
0           1    Male   19                  15                      39
1           2    Male   21                  15                      81
2           3  Female   20                  16                       6
3           4  Female   23                  16                      77
4           5  Female   31                  17                      40

Last 5 Records:
     CustomerID  Gender  Age  Annual Income (k$)  Spending Score (1-100)
195         196  Female   35                 120                      79
196         197  Female   45                 126                      28
197         198    Male   32                 126                      74
198         199    Male   32                 137                      18
199         200    Male   30                 137                      83

Shape of Dataset:
(200, 5)

Data Types:
CustomerID                 int64
Gender                    object
Age                        int64
Annual Income (k$)         int64
Spending Score (1-100)     int64
dtype: object

Missing Values:
CustomerID                0
Gender                    0
Age                       0
Annual Income (k$)        0
Spending Score (1-100)    0
dtype: int64

Number of Duplicate Records:
0

Summary Statistics:
       CustomerID         Age  Annual Income (k$)  Spending Score (1-100)
count  200.000000  200.000000          200.000000              200.000000
mean   100.500000   38.850000           60.560000               50.200000
std     57.879185   13.969007           26.264721               25.823522
```

```
min      1.000000   18.000000          15.000000               1.000000
25%     50.750000   28.750000          41.500000              34.750000
50%    100.500000   36.000000          61.500000              50.000000
75%    150.250000   49.000000          78.000000              73.000000
max    200.000000   70.000000         137.000000              99.000000
```

```
X = df[["Annual Income (k$)", "Spending Score (1-100)"]]

print("Selected Features:")
print(X.head())
```

```
Selected Features:
   Annual Income (k$)  Spending Score (1-100)
0                  15                      39
1                  15                      81
2                  16                       6
3                  16                      77
4                  17                      40
```

```
scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)

print("Standardized Features:")
print(X_scaled[:5])
```

```
Standardized Features:
[[-1.73899919 -0.43480148]
 [-1.73899919  1.19570407]
 [-1.70082976 -1.71591298]
 [-1.70082976  1.04041783]
 [-1.66266033 -0.39597992]]
```

```
plt.figure(figsize=(8, 5))

plt.scatter(
    X["Annual Income (k$)"],
    X["Spending Score (1-100)"]
)

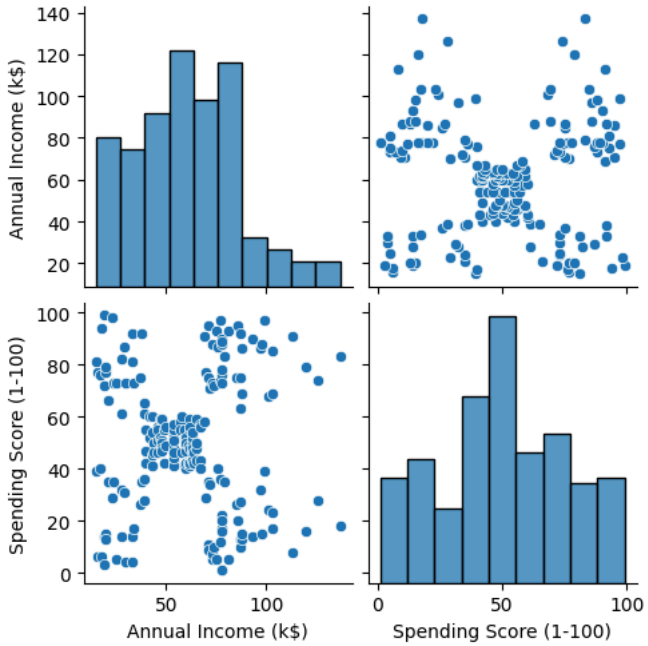
plt.title("Annual Income vs Spending Score")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")

plt.show()


sns.pairplot(X)

plt.show()
```






```
wcss = []

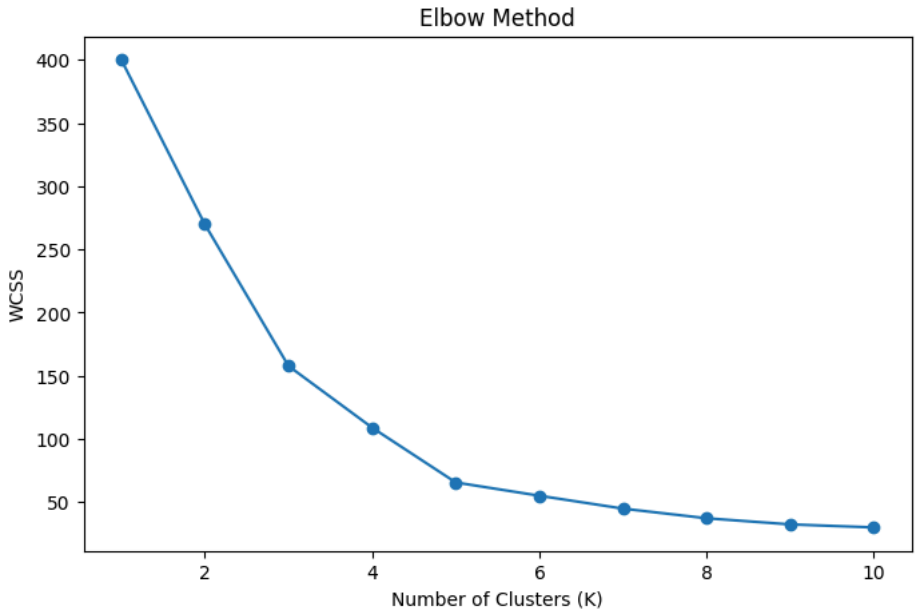
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(8, 5))

plt.plot(range(1, 11), wcss, marker='o')

plt.title("Elbow Method")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")

plt.show()
```




```
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)

df["Cluster"] = kmeans.fit_predict(X_scaled)

print("Cluster Labels:")
print(df[["Annual Income (k$)", "Spending Score (1-100)", "Cluster"]].head())

# Save the dataset
df.to_csv("/content/Mall_Customers_Clustered.csv", index=False)

print("\nClustered dataset saved successfully!")
```

```
Cluster Labels:
   Annual Income (k$)  Spending Score (1-100)  Cluster
0                  15                      39        4
1                  15                      81        2
2                  16                       6        4
3                  16                      77        2
4                  17                      40        4

Clustered dataset saved successfully!
```

```
plt.figure(figsize=(10, 6))

sns.scatterplot(
    data=df,
    x="Annual Income (k$)",
    y="Spending Score (1-100)",
    hue="Cluster",
    palette="viridis",
    s=80
)

centroids = scaler.inverse_transform(kmeans.cluster_centers_)

plt.scatter(
    centroids[:, 0],
    centroids[:, 1],
    s=250,
    c="red",
    marker="X",
    label="Centroids"
)

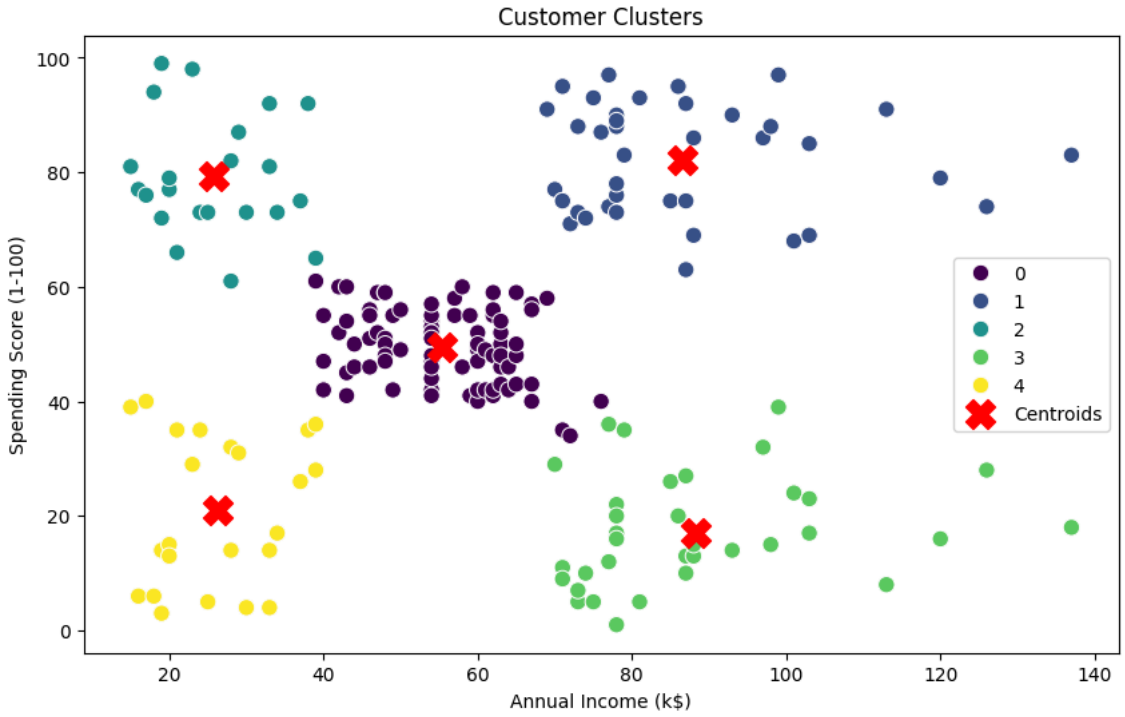
plt.title("Customer Clusters")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.legend()

plt.show()

print("Cluster Centroid Coordinates:")

for i, centroid in enumerate(centroids):
```

```
print(
    f"Cluster {i}: "
    f"Annual Income = {centroid[0]:.2f}, "
    f"Spending Score = {centroid[1]:.2f}"
)
```




```
Cluster Centroid Coordinates:
Cluster 0: Annual Income = 55.30, Spending Score = 49.52
Cluster 1: Annual Income = 86.54, Spending Score = 82.13
Cluster 2: Annual Income = 25.73, Spending Score = 79.36
Cluster 3: Annual Income = 88.20, Spending Score = 17.11
Cluster 4: Annual Income = 26.30, Spending Score = 20.91
```

```
cluster_counts = df["Cluster"].value_counts().sort_index()

print("Number of Customers in Each Cluster:")
print(cluster_counts)
```

```
Number of Customers in Each Cluster:
Cluster
```